

Module/Pathway Review — Flagellar assembly & type III export in *Pseudomonas putida* KT2440 (KEGG ppu02040)

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Bucket:** kegg:ppu02040 "Flagellar assembly" — 47 primary candidate genes **Module area:** transport_motility_signaling **Prepared for:** manual module-satisfiability and gene-annotation curation

1. Executive summary

The KEGG ppu02040 flagellar-assembly module is **fully satisfiable** in *P. putida* KT2440. All six generic module steps — (1) late transcription control, (2) flagellar type III export, (3) MS-ring/switch, (4) rod-hook + length control, (5) filament junction/polymerization/cap, (6) stator/torque — are represented by high-confidence structural genes, most within one contiguous chromosomal region (~PP_4335–PP_4396, the canonical *Pseudomonas* flagellar Region I), with motor and accessory loci distributed elsewhere. KT2440 is polarly flagellated and motile, and its master flagellar regulator **FleQ** is directly validated (ChIP-seq and $\Delta fleQ$ non-motility, target-strain evidence).

Of the 47 candidates, **~39 are solid core flagellar genes**. Curation attention should concentrate on eight entries:

- **Three likely over-annotations to remove** from the flagellar bucket: `fliY` (PP_0227, a cystine ABC-transporter binding protein — a name collision), `PP_5157` (amino-acid ABC binding protein), and `PP_1087` (generic OmpA porin).
- **One generic inclusion to drop:** `rpoD`/ σ^{70} (PP_0387) is housekeeping, not flagellar-specific.
- **One mis-annotation to correct:** `atoC` (PP_4371) is almost certainly **fleR**, the σ^{54} -dependent flagellar response-regulator.

- **Paralog ambiguities to resolve:** the second stator pair PP_4335 / PP_4336 (MotCD/MotY-type) and the second FliL paralog PP_5209.

The only genuine **coverage gaps** are in the regulatory step: **fleS, fleR, and fleN** — dedicated *Pseudomonas* flagellar regulators — are not captured by the bucket (though fleR is likely mislabeled as atoC). The structural pathway itself has no gaps.

Confidence: Structural completeness — high (direct genomic + strong *Pseudomonas* homology). FleQ function — high (direct KT2440 experiments). Regulator identities (atoC→fleR; second stator ortholog) — moderate, flagged for fetch-gene confirmation.

2. Target-organism pathway definition

Process included. Assembly of the bacterial flagellum: the membrane-embedded **flagellum-specific type III secretion system (fT3SS) export apparatus** (FlhA, FlhB, FliP, FliQ, FliR, FliO + cytoplasmic ATPase complex FliI/FliH/FliJ), the **MS-ring** (FliF) and **C-ring/switch** (FliG, FliM, FliN), the **rod** (FlgB, FlgC, FlgF, FlgG, FliE) with rod/hook caps (FlgD, FlgJ), **P- and L-rings** (FlgI/FlgA, FlgH), the **hook** (FlgE) and hook-length control (FliK), the **hook-filament junction** (FlgK, FlgL), **flagellin filament** (FliC) and **cap** (FliD), export **chaperones** (FliS, FliT, FlgN), the **stator/torque units** (MotA/MotB and a second MotCD-type set) plus **FliL**, and the **dedicated transcriptional circuit** (FleQ, FleSR, FliA/ σ^{28} , FlgM anti- σ , FleN, with σ^{54} /RpoN).

Neighboring pathways to keep separate. - **Bacterial chemotaxis** (KEGG ppu02030): Che proteins signal to the C-ring switch (FliM/FliN/FliG) but are a distinct map — do not fold Che genes into ppu02040. - **Two-component systems** (ppu02020): FleS/FleR belong functionally here as a flagellar TCS, but generic TCS/ σ^{70} housekeeping should not be pulled in. - **Bacterial secretion / injectisome T3SS** (ppu03070): evolutionarily homologous to the flagellar export gate, but KT2440 (an environmental saprophyte) lacks a virulence injectisome — the flagellar apparatus is its principal type III secretion system. No separate injectisome module is expected.

Alternate names / database definitions. KEGG map02040 = "Flagellar assembly." GO anchors: flagellum assembly (GO:0009296 / GO:0044780), bacterial-type flagellum (GO:0009288), flagellum-dependent cell motility (GO:0071973), protein secretion by the type III secretion system (GO:0030254). *Pseudomonas* literature refers to the "*fla* regulon" organized in non-contiguous chromosomal regions (Dasgupta 2003,

).

P 14617143

3. Expected step model (satisfiability verdict)

#	Module step	Key players expected	KT2440 candidates	Verdict
1	Late transcription control / regulatory circuit	FleQ, FleS, FleR, FliA(σ^{28}), FlgM, FleN, RpoN(σ^{54})	fleQ (PP_4373), fliA (PP_4341), flgM (PP_4395), rpoN (PP_0952); atoC=likely fleR (PP_4371)	covered but module_needs_revision (add fleS/fleN; fix atoC→fleR; drop rpoD)
2	Flagellar type III export	FlhA, FlhB, FliP, FliQ, FliR, FliO, FliI, FliH, FliJ	all 9 present (PP_4344–PP_4367)	covered
3	MS-ring + C-ring switch	FliF, FliG, FliM, FliN	all present (PP_4357–PP_4369)	covered
4	Rod, rings, hook, length control	FlgB/C/F/G, FliE, FlgD, FlgJ, FlgA, FlgH, FlgI, FlgE, FliK	all present (PP_4361–PP_4394)	covered
5	Junction, filament, cap, chaperones	FlgK, FlgL, FliC, FliD, FliS, FliT, FlgN	all present (PP_4374–PP_4396)	covered
6	Stator installation / torque + FliL	MotA, MotB (+ MotCD set), FliL	motA/motB (PP_4905/PP_4904), PP_4335/PP_4336 (2nd stator), fliL (PP_4359), PP_5209 (2nd FliL)	covered; candidate_uncertain on paralog identity

No step is `not_expected_in_target_taxon`. No structural step is a `gap`.

4. Candidate genes and evidence

Regulatory (step 1). - **fleQ (PP_4373)** — master flagellar/biofilm regulator; **direct KT2440 evidence:** 103 FleQ ChIP-seq binding sites (Blanco-Romero 2018,

P 30177764

ΔfleQ is non-motile with reduced biofilm (Kim 2024,

P 39570920

High confidence. - **fliA (PP_4341)** σ^{28} and **flgM (PP_4395)** anti- σ^{28} — canonical late-gene checkpoint; homology + operon context; high confidence. - **rpoN (PP_0952)** σ^{54} — required for Class II flagellar transcription in *Pseudomonas*

P 14617143

shared with many σ^{54} regulons (not flagellar-exclusive). Keep, with caveat. - **atoC (PP_4371)** — annotated generic σ^{54} -dependent NtrC/AtoC activator; located inside the flagellar operon between the *fli* genes and *fleQ*; **almost certainly FleR** (the flagellar σ^{54} response regulator). Curation-relevant mis-name → **promote to fetch-gene**. - **rpoD (PP_0387)** σ^{70} — housekeeping sigma; generic KEGG-map artifact, **not flagellar-specific**.

Export apparatus (step 2). flhA, flhB, fliP, fliQ, fliR, fliO, fliI, fliH, fliJ — complete membrane export gate + cytoplasmic ATPase (FliI, EC 7.1.2.2). Note the EC on FliI is a broad F₁-ATPase-like mapping; export can also be PMF-driven via FlhA/FlhB, so treat the EC as descriptive, not a distinct enzymatic step. High confidence by strong *Pseudomonas*/enterobacterial homology + tight operon synteny.

Basal body / switch / rod / hook / rings / filament / chaperones (steps 3–5). fliF, fliG, fliM, fliN; flgB/C/F/G, fliE, flgD, flgJ, flgA, flgH, flgI, flgE, fliK; flgK, flgL, fliC, fliD, fliS, fliT, flgN (PP_4396). All are core, unambiguous, syntenic within Region I; high confidence. Caveat: FliO is poorly conserved and short — low sequence identity is expected and should not be read as absence.

Stator/torque + FliL (step 6). - **motA/motB (PP_4905/PP_4904)** and **PP_4335/PP_4336** = two homologous stator sets, matching the **MotAB + MotCD** architecture of *Pseudomonas* (Zhang 2022,

P 36286538

Transfer to KT2440 is strong (close relative, conserved synteny). Ortholog assignment of which pair is MotAB vs MotCD (and whether MotX/MotY-type components exist) needs fetch-gene. -

fliL (PP_4359) and PP_5209 (2nd FliL) — FliL rings the stator to tune torque/switching and stabilizes the rod (Partridge 2024 review

P 39096095

P. aeruginosa

P 36286538

Two paralogs likely partition between the two stator systems.

5. Gaps, ambiguities, and likely over-annotations

Likely over-annotations (recommend removal from ppu02040): - **fliY (PP_0227)** — annotated "Periplasmic cystine-binding protein." Enterobacterial "FliY" is the substrate-binding subunit of the **FliY/YecS/YecC cystine ABC importer** (Deutch 2014,

P 25139244

not a flagellar C-ring protein. Proteobacteria use the FliM+FliN switch and lack the Bacillus FliM/FliN/FliY switch, so a flagellar FliY is not expected. Classic KEGG **name-collision over-propagation**. - **PP_5157** — "Solute-binding protein family 3": another amino-acid ABC substrate-binding protein; no flagellar role. - **PP_1087** — generic OmpA-family outer-membrane porin; not a documented flagellar component (the OmpA/peptidoglycan-binding domain it shares with MotB is a domain-level, not functional, match).

Generic inclusion to drop: rpoD/ σ^{70} (PP_0387) — housekeeping.

Ambiguity / mis-annotation: atoC (PP_4371) → FleR (see §4).

Genuine coverage gaps (regulatory step): fleS (sensor kinase) and **fleN** (FleQ anti-activator P-loop ATPase controlling flagellar number; Jain 2016,

P 26841764

are expected in the KT2440 genome but **not captured** by the bucket. **fleR** is likely present but mislabeled as atoC. → mark step 1 **module_needs_revision**.

Not over-annotation but worth noting: two flagellin/HAP paralogs and possible flagellin glycosylation islands vary across *Pseudomonas* (P 14617143);
if a second flagellin exists in KT2440 it would be an addition, not a gap.

6. Module and GO-curation recommendations

Step-level status: - Steps 2, 3, 4, 5: **covered**. - Step 6 (stator): **covered**, but set the two extra stator genes and PP_5209 to **candidate_uncertain** pending ortholog assignment. - Step 1 (regulation): **covered but module_needs_revision** — add fleS and fleN; reassign atoC→fleR; drop rpoD as non-specific.

Bucket hygiene: - Remove fliY (PP_0227), PP_5157, PP_1087, and rpoD (PP_0387) from the flagellar satisfiability count so they do not inflate coverage. - Keep chemotaxis (ppu02030), generic TCS (ppu02020), and injectisome T3SS (ppu03070) as **separate** modules; mark injectisome **not_expected_in_target_taxon**.

Module-boundary judgement: the generic six-step boundary is essentially **correct** for KT2440. One refinement: the regulatory step should explicitly enumerate the *Pseudomonas* enhancer-binding-protein circuit (FleQ–FleSR–FleN– σ^{54} + FliA/FlgM– σ^{28}) rather than the enterobacterial FlhDC master operon, which *Pseudomonas* **does not use** — a lineage-specific alternative worth encoding in the module document.

GO/term requests: existing GO terms (GO:0044780 flagellum assembly, GO:0071973 flagellum-dependent motility, GO:0030254 T3SS protein secretion, GO:0006935 chemotaxis for the neighbor) are sufficient; no new terms needed. A curator note recording the FleQ-master-regulator (vs FlhDC) lineage distinction would prevent future over-propagation.

7. Genes to promote to full fetch-gene review

1. **PP_4371 (atoC → FleR):** confirm identity as the flagellar σ^{54} response regulator; highest curation impact.
2. **PP_4335 / PP_4336:** assign as MotCD (or MotAB) stator set; distinguish from PP_4904/PP_4905.

3. **PP_5209 (2nd FliL)**: confirm paralog and its stator partner.
 4. **PP_0227 (fliY)**: confirm reclassification as cystine-ABC binding protein and removal from the flagellar bucket.
 5. **PP_5157 / PP_1087**: confirm non-flagellar (ABC binding protein / OmpA porin) and remove.
 6. (Add if present in genome) **fleS, fleN** loci: locate and add to the regulatory step.
-

8. Key references

- Blanco-Romero E, *et al.* (2018) Genome-wide analysis of the FleQ direct regulon in *Pseudomonas fluorescens* F113 and *Pseudomonas putida* KT2440.

P 30177764

— *direct KT2440 evidence for FleQ as master regulator (103 ChIP-seq sites).*

- Kim, Lee, Darlington, Kim (2024) Impact of *fleQ* deficiency on resource allocation and heterologous gene expression in *Pseudomonas putida*.

P 39570920

— *ΔfleQ KT2440 is non-motile.*

- Dasgupta N, *et al.* (2003) A four-tiered transcriptional regulatory circuit controls flagellar biogenesis in *Pseudomonas aeruginosa*.

P 14617143

— *defines FleQ/FleS/FleR/FliA/FlgM/FleN + RpoN circuit and non-contiguous flagellar regions (strong transfer to KT2440).*

- Jain (2016) Crystallization/analysis of FleN from *P. aeruginosa*.

P 26841764

— *FleN is a P-loop ATPase anti-activator of FleQ controlling flagellar number.*

- Zhang C, *et al.* (2022) FliL differentially interacts with two stator systems to regulate flagellar motor output in *Pseudomonas aeruginosa*.

P 36286538

— *two homologous MotAB/MotCD stators + FliL.*

- Partridge JD, Harshey RM (2024) Flagellar protein FliL: a many-splendored thing.

P 39096095

— *FliL functions at the stator/rod.*

- Deutch CE, *et al.* (2014) L-selenaproline uptake depends on two L-cystine transport systems.

P 25139244

— *FliY is the cystine ABC-transporter binding protein (supports over-annotation call).*

Evidence provenance summary

- **Direct KT2440 experimental:** FleQ master-regulator role

(

P 30177764

).

P 39570920

- **Strong *Pseudomonas* transfer:** four-tiered regulatory circuit, FleN, two-stator system

(

P 14617143

P 26841764

).

P 36286538

— close relatives, conserved synteny.

- **Homology / operon-context inference:** all structural export/basal-body/rod-hook/filament genes (Region I synteny + universal bacterial conservation).

- **Over-annotation calls:** fliY/PP_5157/PP_1087/rpoD based on family assignment and KEGG map scope

(

P 25139244

for fliY).

Open questions for experts: (i) Does KT2440 encode fleS/fleN at loci outside the bucket, and is PP_4371 confirmed as fleR? (ii) Which stator pair is MotAB vs MotCD, and do accessory MotX/MotY components exist? (iii) Is there a second flagellin / glycosylation island? These are the highest-value resolutions for finalizing module satisfiability.

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery